

Unusual Does Not Mean Impossible

AP Statistics · Unit 2 · Topic 2.10 · B: 2026-11-06 · E: 2026-11-30 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Using Probability and Simulation: Determine parameters for probability distributions.
- **Skill 4.B** — Statistical Argumentation: Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.C** — Calculate parameters for a binomial distribution. [Skill 3.B]
- **EK UNC-3.C.1** — If a random variable is binomial, its mean, μ_X , is np and its standard deviation, σ_X , is $\sqrt{np(1-p)}$.
- **LO UNC-3.D** — Interpret probabilities and parameters for a binomial distribution. [Skill 4.B]
- **EK UNC-3.D.1** — Probabilities and parameters for a binomial distribution should be interpreted using appropriate units and within the context of a specific population or situation.

Essential question: How do binomial parameters help distinguish an unusual count from an impossible one?

DOK-3 skill: 4.B · **FRQ pattern:** binomial-mean-sd-unusual-vs-impossible

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1
→ 3 **Minutes: 45**

Teacher does

- Show the board at the bell and collect a one-sentence first take without confirming a claim.
- Run the named video follow-along, then allow about 10 minutes for parts (a)–(c).
- Ask for evidence linked to a claim. Collect the sheet and hand-score part (c) with E/P/I.

Students do

- Choose the coach or analyst and name one quantity to check before the lesson.
- Complete the follow-along about binomial mean, standard deviation, and interpretation.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

Adult role

Circulate and ask students to connect their choice to evidence. Score part (c) using the three required reasoning elements; do not require dropped extension work.

[WARN] WATCH FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Compares 58 with mean 70 and SD about 4.6 in count units
- **reasoning-2** (required) — Uses the positive 0.0072 chance to distinguish unusual from impossible
- **reasoning-3** (required) — Supports the analyst and explains the coach's false guaranteed cutoff or proof claim

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Using 0.70 as the mean count instead of 70
- Omitting the square root in the standard deviation
- Treating an unusual outcome as impossible or as proof that the model changed

[STAR] TODAY'S PROBLEM — annotated

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: “Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies.”

Analyst: “A result can be unusual without being impossible; compare 58 with the model’s mean and standard deviation.”

The binomial model gives a chance of about 0.0072 for 58 or fewer makes; use this provided value.

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE (5 min)

Whose statement is more defensible, the coach’s or the analyst’s? Name one quantity you would check.

Key: Not graded. Listen for whether students use impossible to mean unlikely; the calculation should sharpen that language.

Rules callout “CENTER, SPREAD, AND UNUSUAL COUNTS (keep on the page)” is printed on the student sheet.

DOK 1 (a) State n and p for this model.

Key: $n = 100$ attempts and $p = 0.70$ for each attempt.

DOK 2 (b) Calculate the binomial mean and standard deviation. Interpret the mean for many sets of 100 shots.

Key: $\mu_X = 100(0.70) = 70$ makes and $\sigma_X = \sqrt{100(0.70)(0.30)} = \sqrt{21} \approx 4.6$ makes. Across many comparable sets of 100 attempts, the model predicts an average of 70 makes.

DOK 3 (c) Whose statement is supported? Compare 58 with the mean and standard deviation, and use the given probability to distinguish unusual from impossible. Explain what the coach’s claim could make a reader wrongly believe. Write 3–4 sentences.

Key: The analyst is supported: 58 is 12 makes below the mean of 70, more than twice the typical distance of about 4.6 makes. The given chance of 58 or fewer is about 0.0072, so such a low result is unusual but has positive probability. The coach turns a typical performance level into a guaranteed lower bound. That could make a reader think one unusual set proves the career rate changed, even though chance can produce it.

EXIT

One thing the video changed about my first take: _____